Problem Sets Before Test 1

Only turn in problems that are ***not*** *bracketed.* Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

notation	meaning
unbracketed	assigned problem – turn these in for grading
()	helper/warm-up problem
[]	additional problems (you are responsible for content, but don't turn them in)
{}	covers optional material

Due	Task	Source	Problems
Wed 2/1	PS 1	Rosen 9.1	(1) (3,5,7,9) types of graph **4** types of graph **8** types of graph **11** graphs and relations **13** intersection graph **18** Fred **31** precedence graph
Fri 2/2	WW01	Rosen 9.2	
Mon 2/6	WW02	Rosen 9.3	
Wed 2/8	PS 2	Rosen 9.4	**2** paths **6** components **12** strong/weak connectivity **18** isomorphic? **20** isomorphic? **54** farmer puzzle [1] paths [3–5] [11] strong/weak connectivity [19, 21] isomorphic?
		Rosen 9.5	**2–8 even** Euler paths/circuits **10** bridges **18** directed Euler **20** directed Euler **26** which have Euler circuits? [1–7 odd] Euler paths/circuits [19, 21] directed Euler Note: Your answers to homework in this class should typically include some sort of explanation. Answers like "Yes", "No", and "17", are rarely sufficient.
Fri 2/9	PS 3	Rosen 9.7	**2–4** planar **12** regions [13] regions **14** vertices
Mon 2/12	PS 4	Rosen 9.RQ	**3** degree
		Rosen 9.SE	**3–5** isomorphic?
Wed 2/14	PS 5	Rosen 9.7	**6–8** planar **23–25** Kuratowski
Fri 2/16	PS 6	Rosen 9.5	**22–23** directed Euler **32–40 even** Hamilton
		Rosen 9.7	**16** bipartite, planar
Mon 2/19	WW03	Rosen 5.1	
Wed 2/21	WW04	Rosen 5.1 – 5.2	

Due	Task	Source	Problems
Fri 2/23	PS 6	Rosen 5.2	**4** bowl **18** students **29** initials **40** houses
		Rosen 5.3	**1** permunations **8** race [11] bits **12** bits [15] letters **18** coins [19] coins **23** line up [24] line up **30** committee
Mon 2/26	WW05	Rosen 5.1–5.4	
Wed 2/28		Rosen 5.RQ	[1–3] counting principles [5] [6–7] pigeonhole principle [8] combinations and permutations [13] counting problem schemas [15] scrambling letters
		Rosen 5.SE	[1-2] selecting items [3] T/F [4] bitstrings [6] phone numbers [7] ice cream [12] zodiac [15] cards [21] donuts [22] permuations [23] combinations [34] round table [35] advising [38] PEPPERCORN [40] license plates
		Rosen 9.RQ	[1] types of graph [2] models [3–5] degree [6–7] graph examples [8] bipartite [9] graph representation [10] isomorphism [11] connected [12] checking isomorphism [13] Euler paths [14a] Hamilton paths [17] planar graph [18] Euler's formula [19] Kuratowski's Theorem
		Rosen 9.SE	[3–5] isomorphic? [21] invariants